

Database Technologies-Assignment 5

Use appropriate SQL functions to solve following queries

Note: To solve below queries, use “hr” database

1. Display all the first names in lower case and the last names in upper case as the single column separated with ' - '. Provide the meaningful name to the column.

```
KD2_98846_Raj>SELECT CONCAT(lower(first_name),'-',upper(last_name)) FROM emp  
loyees;
```

CONCAT(lower(first_name),'-',upper(last_name))
ellen-ABEL
sundar-ANDE
mozhe-ATKINSON
shelli-BAIDA
amit-BANDA
elizabeth-BATES
sarah-BELL
david-BERNSTEIN
laura-BISSOT
harrison-BLOOM
hermann-BROWN
alexis-BULL
anthony-CABRIO
gerald-CAMBRAULT
nanette-CAMBRAULT
john-CHEN
kelly-CHUNG
karen-COLMENARES
curtis-DAVIES
pat-DAVIS
julia-DELLINGER
jennifer-DILLY
louise-DORAN
alberto-ERRAZURIZ
britney-EVERETT
daniel-FAVIET
kevin-FEENEY
jean-FLEAUR
tayler-FOX
adam-FRIPP
lex-GARCIA
ki-GEE

2. Display the first word of the job title. (Hint : Use substring_index).

```
KD2_98846_Raj>SELECT SUBSTRING_INDEX(job_title, ' ', 1) AS first_word
-> FROM jobs;
+-----+
| first_word |
+-----+
| Public     |
| Accounting |
| Administration |
| President  |
| Administration |
| Accountant |
| Finance    |
| Human      |
| Programmer |
| Marketing  |
| Marketing  |
| Public     |
| Purchasing |
| Purchasing |
| Sales      |
| Sales      |
| Shipping   |
| Stock      |
| Stock      |
+-----+
19 rows in set (0.00 sec)
```

3. Display the First name, last name and length of first name for the employees whose last name contains character 'b' anywhere after 3rd position.

```
KD2_98846_Raj>SELECT first_name, last_name, LENGTH(first_name) AS name_length
-> FROM employees
-> WHERE INSTR(SUBSTRING(last_name, 4), 'b') > 0;
+-----+-----+-----+
| first_name | last_name | name_length |
+-----+-----+-----+
| Gerald     | Cambrault | 6           |
| Nanette    | Cambrault | 7           |
| Nancy      | Gruenberg | 5           |
| Susan      | Jacobs    | 5           |
+-----+-----+-----+
4 rows in set (0.01 sec)

KD2_98846_Raj>
```

4. Display the employee id, first name and last name of the employees whose first or last name includes a space in between.

```
KD2_98846_Raj>SELECT employee_id , first_name , last_name FROM employees WHERE first_name LIKE '% %'OR last_name LIKE '% %';
```

employee_id	first_name	last_name
112	Jose Manuel	Urman

```
1 row in set (0.00 sec)
```

5. Display first name, salary, and round the salary to thousands.

```
KD2_98846_Raj>SELECT first_name, salary, ROUND(salary, -3) AS rounded_salary
-> FROM employees;
```

first_name	salary	rounded_salary
Steven	24000.00	24000
Neena	17000.00	17000
Lex	17000.00	17000
Alexander	9000.00	9000
Bruce	6000.00	6000
David	4800.00	5000
Valli	4800.00	5000
Diana	4200.00	4000
Nancy	12008.00	12000
Daniel	9000.00	9000
John	8200.00	8000
Ismael	7700.00	8000
Jose Manuel	7800.00	8000
Luis	6900.00	7000
Den	11000.00	11000
Alexander	3100.00	3000
Shelli	2900.00	3000
Sigal	2800.00	3000
Guy	2600.00	3000
Karen	2500.00	3000
Matthew	8000.00	8000

6. Display employee ID and the date on which he ended his previous job in the format as mentioned.
(Format: Thursday, October 27th 2011)

```
KD2_98846_Raj>SELECT EMPLOYEE_ID,DATE_FORMAT(END_DATE,'%W,%M %D %Y') AS formatted_date FROM job_history;
```

EMPLOYEE_ID	formatted_date
101	Thursday, October 27th 2011
101	Sunday, March 15th 2015
102	Sunday, July 24th 2016
114	Sunday, December 31st 2017
122	Sunday, December 31st 2017
176	Saturday, December 31st 2016
176	Sunday, December 31st 2017
200	Friday, June 17th 2011
200	Saturday, December 31st 2016
201	Tuesday, December 19th 2017

10 rows in set (0.04 sec)

7. Display first name and date of first salary of the employees. (Consider 1st day of month as salary day.
HINT: LAST_DAY).

```
KD2_98846_Raj>SELECT first_name,
->      DATE_SUB(hire_date, INTERVAL DAY(hire_date) - 1 DAY) AS first_salary_date
-> FROM employees;
```

first_name	first_salary_date
Steven	2013-06-01
Neena	2015-09-01
Lex	2011-01-01
Alexander	2016-01-01
Bruce	2017-05-01
David	2015-06-01
Valli	2016-02-01
Diana	2017-02-01
Nancy	2012-08-01
Daniel	2012-08-01
John	2015-09-01
Ismael	2015-09-01
Jose Manuel	2016-03-01
Luis	2017-12-01
Den	2012-12-01
Alexander	2013-05-01
Shelli	2015-12-01
Sigal	2015-07-01
Guy	2016-11-01
Karen	2017-08-01
Matthew	2014-07-01
Adam	2015-04-01
Payam	2013-05-01
Shanta	2015-10-01
Kevin	2017-11-01
Julia	2015-07-01
Irene	2016-09-01
James	2017-01-01
Steven	2018-03-01
Laura	2015-08-01
Mozhe	2015-10-01

8. Display first name and experience of the employees in the format of number of years, months and days completed.

```
KD2_98846_Raj>SELECT first_name,
->      CONCAT(
->        TIMESTAMPTDIFF(YEAR, hire_date, CURDATE()), ' years ',
->        MOD(TIMESTAMPTDIFF(MONTH, hire_date, CURDATE()), 12), ' months ',
->        DATEDIFF(
->          CURDATE(),
->          DATE_ADD(
->            DATE_ADD(
->              hire_date,
->              INTERVAL TIMESTAMPTDIFF(YEAR, hire_date, CURDATE()) YEAR
->            ),
->            INTERVAL MOD(TIMESTAMPTDIFF(MONTH, hire_date, CURDATE()), 12) MONTH
->          ), ' days'
->      ) AS experience
-> FROM employees;
```

first_name	experience
Steven	13 years 2 months 21 days
Neena	10 years 11 months 17 days
Lex	15 years 7 months 25 days
Alexander	10 years 8 months 4 days
Bruce	9 years 3 months 17 days
David	11 years 2 months 13 days
Valli	10 years 7 months 2 days
Diana	9 years 7 months 0 days
Nancy	14 years 0 months 21 days
Daniel	14 years 0 months 22 days
John	10 years 11 months 10 days
Ismael	10 years 11 months 8 days
Jose Manuel	10 years 6 months 0 days
Luis	8 years 9 months 0 days
Den	13 years 9 months 0 days
Alexander	13 years 3 months 20 days
Shelli	10 years 8 months 14 days
Sigal	11 years 1 months 14 days
Guy	9 years 9 months 23 days
Karen	9 years 0 months 28 days
Matthew	12 years 1 months 20 days
Adam	11 years 4 months 28 days
Payam	13 years 4 months 6 days
Shanta	10 years 10 months 28 days
Kevin	8 years 0 months 23 days

9. Display the employee id, first name and joining date of the employees who joined in the month of august of any year.

```
KD2_98846_Raj>SELECT employee_id, first_name, Hire_date FROM EMPLOYEES WHERE MONTH(Hire_date) = 8;
```

employee_id	first_name	Hire_date
108	Nancy	2012-08-17
109	Daniel	2012-08-16
119	Karen	2017-08-10
129	Laura	2015-08-20
134	Michael	2016-08-26
152	Peter	2015-08-20
158	Allan	2014-08-01
189	Jennifer	2015-08-13
202	Pat	2015-08-17

```
9 rows in set (0.00 sec)
```

10. Display the details of employees who joined before the year 2015.

```
KD2_98846_Raj>SELECT * FROM employees WHERE Hire_date < '2015-01-01';
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID
SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID			
100	Steven	King	SKING	1.515.555.0100	2013-06-17	AD_PRES
24000.00		NULL	NULL	90		
102	Lex	Garcia	LGARCIA	1.515.555.0102	2011-01-13	AD_VP
17000.00		NULL	100	90		
108	Nancy	Gruenberg	NGRUENBE	1.515.555.0108	2012-08-17	FI_MGR
12008.00		NULL	101	100		
109	Daniel	Faviet	DFAVIET	1.515.555.0109	2012-08-16	FI_ACCOUNT
9000.00		NULL	108	100		
114	Den	Li	DLI	1.515.555.0114	2012-12-07	PU_MAN
11000.00		NULL	100	30		
115	Alexander	Khoo	AKHOO	1.515.555.0115	2013-05-18	PU_CLERK
3100.00		NULL	114	30		
120	Matthew	Weiss	MWEISS	1.650.555.0120	2014-07-18	ST_MAN
8000.00		NULL	100	50		
122	Payam	Kaufling	PKAUFLIN	1.650.555.0122	2013-05-01	ST_MAN
7900.00		NULL	100	50		
133	Jason	Mallin	JMALLIN	1.650.555.0133	2014-06-14	ST_CLERK
3300.00		NULL	122	50		
137	Renske	Ladwig	RLADWIG	1.650.555.0137	2013-07-14	ST_CLERK
3600.00		NULL	123	50		
141	Trenna	Rajs	TRAJS	1.650.555.0141	2013-10-17	ST_CLERK
3500.00		NULL	124	50		
145	John	Singh	JSINGH	44.1632.960000	2014-10-01	SA_MAN
14000.00		0.40	100	80		
156	Janette	King	JKING	44.1632.960011	2014-01-30	SA_REP
10000.00		0.35	146	80		
157	Patrick	Sully	PSULLY	44.1632.960012	2014-03-04	SA_REP
9500.00		0.35	146	80		
158	Allan	McEwen	AMCEWEN	44.1632.960013	2014-08-01	SA_REP
9000.00		0.35	146	80		
174	Ellen	Abel	EABEL	44.1632.960029	2014-05-11	SA_REP
11000.00		0.30	149	80		
184	Nandita	Sarchand	NSARCHAN	1.650.555.0149	2014-01-27	SH_CLERK
4200.00		NULL	121	50		
192	Sarah	Bell	SBELL	1.650.555.0157	2014-02-04	SH_CLERK
4000.00		NULL	123	50		

11. Display the number of days between system date and 1st January 2011.

```
KD2_98846_Raj>SELECT DATEDIFF(CURDATE(), '2011-01-01') AS number_of_days;
```

number_of_days
5728

1 row in set (0.00 sec)

12. Display number of employees joined after 15th of any month.

```
KD2_98846_Raj>SELECT COUNT(*) AS employee_count  
-> FROM employees  
-> WHERE DAY(hire_date) > 15;
```

```
+-----+  
| employee_count |  
+-----+  
|          57 |  
+-----+  
1 row in set (0.01 sec)
```

13. Display third highest salary of employees.

```
KD2_98846_Raj>SELECT DISTINCT salary  
-> FROM employees  
-> ORDER BY salary DESC  
-> LIMIT 1 OFFSET 2;
```

```
+-----+  
| salary |  
+-----+  
| 14000.00 |  
+-----+  
1 row in set (0.01 sec)
```